Swarm Witness

An executable evidence protocol for agent authorization claims — binding claims to observations and reporting support, refutation, or unresolved.

Jonathan Maye-Hobbs · Independent
With Apart Research

PROBLEM

Passing tests ≠ warranted claims

- Agents present receipts, copied grants, and restart traces that look convincing.
- Subsystems can pass while the alleged right to act remains unsupported.
- Delegation makes attribution and *current* authority evidence questions.

Question: What evidence is sufficient to justify an authorization claim about a decision or effect?

CONTRIBUTION

Qualify claims against observations

Protocol

- Typed Claim / Run / Finding records
- Separate evidence, decision, effect, comparison axes
- Supported · Refuted · Unresolved

Controls

- Source-pinned re-execution (R1)
- Later-harness confirmation (R2)
- Native BMO vs SQLite (R3-L1)
- Causal multi-principal schedules + formal checker

A test can pass while the claim stays unwarranted

R3-D worked example

- **Test:** ClassicalGate completes 4 shared cases with expected decisions
- **Claim:** matched decision-model equivalence with BMO
- **Missing:** isomorphic BMO observations (0/4)
- **Verdict:** unresolved / missing_evidence

What this is not

Not a negative BMO result. Not affinity support. R4 `passed: true` only means expected classifier labels —including missing-evidence.

MEASURED OUTCOMES

Designed schedules, separated counts

9 / 0

Unauthorized effects · faulty local vs reference (R5 v2)

14 scenarios × 2 policies · 10 incorrect decisions under faulty policy (includes one false denial with no commit)

6 / 6

R3-L1 first-edit cases matching in both arms

Plus R1/R2 pinned confirmations · 2,646 finite formal checks · receipt± isolation without local-view confound

Identical grant observations ≠ current permission

World	Cached view	Authority	Reference	Faulty local	Unauthorized effects
A, valid	Old active snapshot	Active	Allow	Allow	0
B, revoked	Same snapshot	Revoked	Deny	Allow	1
B + irrelevant	Same snapshot	Revoked	Deny	Allow	1

Receipt-present vs receipt-absent (no qualifying local view for C) isolates the receipt rule itself.

Three-way evidence judgment

- Compatible worlds $\Gamma(o)$ under authenticated observations
- Support when $\Gamma(o) \subseteq P$ · Refutation when $\Gamma(o) \cap P = \varnothing$ · else unresolved
- Incomplete evidence widens Γ — may require unresolved; does not by itself invalidate soundness
- Falsified observations or an unsound compatibility model can invalidate premises

Proofs cover the specified set semantics; the audit evaluator is assessed on the declared scenario corpus.

TAKEAWAY

Make the warrant gap mechanically visible

- Successful execution, selected-observation agreement, and support for a stated property are distinct outcomes.
- Swarm Witness leaves claims unresolved when required observations are absent.
- Classical reference-monitor behavior is the null control — not a new permission primitive.

Code	github.com/instagrim-dev/swarm-witness
Bundle	dist/swarm-witness-review-bundle.zip
Report	Templated PDF + research PDF under report/